

pyMack: Validation and Verification

An agreement audit of the solver against published benchmarks
at their exact stated conditions

pyMack technical documentation

July 2, 2026

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1 Scope and methodology

This document is the agreement audit for pyMack, the compressible linear-stability solver whose formulation and numerics are developed in the companion documents (*The Mean Boundary-Layer Profile, Derivation of the Disturbance Equations, Numerical Methods*). Each case below (i) reruns a published or external reference at its *exact* stated conditions, (ii) measures agreement against the digitised or tabulated reference, and (iii) records a verdict on a fixed three-tier scale applied uniformly to every case,

$$e \leq 5\% \Rightarrow \text{agrees}, \quad 5\% < e \leq 15\% \Rightarrow \text{acceptable}, \quad e > 15\% \Rightarrow \text{disagrees}, \quad (1)$$

where e is the case’s headline relative error; dimensional cases use the mean absolute error as a fraction of the curve span. Nothing is tuned to pass, and the disagreements are kept in full view.

Organisation: an evidence pyramid. The cases are ordered from the most rigorously anchored physics to the engineering configurations, each tier earning trust only once the tier beneath it holds:

1. **Incompressible core.** The Orr–Sommerfeld temporal eigenvalue for plane Poiseuille flow [1] is a scalar with a textbook-exact answer. Matching it to $\sim 10^{-5}$ confirms the spectral discretisation, the eigenvalue assembly, and the boundary-condition enforcement in isolation, before any compressible physics is added.
2. **Compressible eigenvalue anchors.** Malik’s tabulated compressible eigenvalues [2] and the Balakumar–Malik discrete/continuous-spectrum topology [3] pin the compressible operator to published digits at single (R, ω) stations; these are the sharpest possible tests, free of digitisation error.
3. **Curve families.** Mack’s neutral curves and growth-rate families [4], the Ma–Zhong $M=4.5$ neutral curve [7], and the Özgen–Kırcalı Fig. 3 neutral-curve family [5] test full *curves* over Reynolds number and Mach number and test topology as well as magnitude. They are reported by physical mode: second mode (Section 4), first mode (Section 5), and the Özgen family, whose panels span *both* modes, separately (Section 6).
4. **Engineering and open cases.** A Mach-6 flat plate [6] anchors the second mode on a flight-relevant configuration. The remaining engineering cases (a sharp-cone N -factor, an independent collaborator’s Mach 5.35 code, and the spatial N -factor pipeline) carry unfinished elements and, together with the $M \approx 2.2$ first-mode disagreements, are collected in the companion note `work_in_progress.tex`.

The base of the pyramid is exact; the middle is a mix of exact (second mode) and partial (first mode). The second-mode design target is trustworthy throughout; the first mode is usable with the documented caveats.

Provenance and reproducibility. All figures embedded here are `pyMack`’s own overlays, `pyMack` curves plotted over digitised reference *points*, so no copyrighted reference figure is reproduced. Every number and verdict is taken verbatim from the committed per-case `verdict.json` files under `verification/` and the repository success matrix, which are regenerated by the scripts recorded in each case directory. This document reports only the settled validations; the open items are tracked in `work_in_progress.tex`.

1.1 Summary of all cases

Table 1 lists every case presented below, in pyramid order; the one-line headline gives the governing number, and the caveats live in each subsection. The Malik temporal test cases 3, 4, and 5, previously logged non-reproducible for want of their dimensional conditions, are now run and pass, once Malik’s own Table I supplied the missing T_0 and wall temperatures; the only remaining *pending* companions are Malik case 1 (the effectively-incompressible $M = 0.5$ anchor, scoped to the Orszag tier) and the Mack Fig. 10.4/10.6 family aggregates, named in prose rather than tabulated.

Table 1: Cases presented in this document, in pyramid order. Verdict scale: Eq. (1) ($\leq 5\%$ *agrees*, 5–15% *acceptable*, else *disagrees*; dimensional cases use MAE as a fraction of curve span).

Case	Source	Quantity	Verdict
<i>Incompressible core</i>			
Plane Poiseuille	Orszag [1]	O–S eigenvalue c	agrees ($\sim 10^{-5}$)
OS spectrum (32 modes)	Orszag [1], Tab. 5	full temporal spectrum	agrees ($< 10^{-5}$)
<i>Compressible eigenvalue anchors</i>			
Malik case 6	Malik [2], Tab. IX	spatial α (2nd mode)	agrees
Malik cases 3/4/5	Malik [2], Tab. IV/V/VI	temporal ω (M2.5/10)	agrees
Malik Table X	Malik [2], Tab. X	spatial α (M10 2nd mode)	acceptable
Malik Fig. 4 eigenfn.	Malik [2], Fig. 4	M10 2nd-mode $\hat{T}(y)$ shape	agrees
B&M branches	Balakumar–Malik [3]	discrete/continuous topology	acceptable
B&M via Xi–Ren–Fu	Balakumar–Malik [3]	spatial α (2nd mode)	acceptable
<i>Second mode: curves & families</i>			
Mack Fig. 10.6	Mack [4]	$\omega_{i,\max}(R)$, M4.5–10	agrees / acceptable
Egorov M6	Egorov et al. [6]	most-amplified ω	acceptable
Ma & Zhong M4.5	Ma & Zhong [7]	neutral curve, spatial 2nd mode	agrees
<i>First mode: curves & families</i>			
Mack Fig. 10.1	Mack [4]	neutral loop, M1.6	acceptable
Mack Fig. 10.3	Mack [4]	$\omega_{i,\max}(R)$, M1.3/3.0	agrees
Mack Fig. 10.4	Mack [4]	oblique 1st mode, M4.5–10	agrees / acceptable
<i>Mixed first + second mode</i>			
Özgen Fig. 3	Özgen–Kırçalı [5]	neutral-curve family (both modes), M2–10	agrees / acceptable

2 Incompressible core

2.1 Orszag (1971): plane-Poiseuille Orr–Sommerfeld eigenvalue

Reference. Orszag [1], the canonical plane-Poiseuille temporal eigenvalue.

Conditions. Plane Poiseuille flow $U = 1 - y^2$ on $[-1, 1]$, Reynolds number $Re = 10\,000$, stream-wise wavenumber $\alpha = 1$, temporal formulation.

Quantity. The least-stable eigenvalue (phase speed) $c = c_r + ic_i$.

Result: agrees. `pyMack`’s Chebyshev Orr–Sommerfeld solver (`validation/test_orszag_full_spectrum.py`) reproduces the benchmark $c = 0.23753 + 0.00374i$ to an absolute error below 10^{-4} already at $N = 64$, tightening toward $\sim 10^{-7}$ under refinement (Table 2). Because this is a scalar comparison against an essentially exact reference rather than a hand-digitised curve, no overlay figure is warranted. The same test module also recovers the Blasius Tollmien–Schlichting mode at $Re_{\delta^*} = 998$, $\alpha = 0.179$ ($c_r \approx 0.36$, unstable) and finds unstable growth at $Re = 1000$, confirming the mapped semi-infinite domain and the growth-rate optimisation loop; these are sign/range assertions rather than digit-exact anchors and are not tabulated separately.

Table 2: Plane Poiseuille temporal eigenvalue, $Re = 10\,000$, $\alpha = 1$. Target $c = 0.23753 + 0.00374i$ [1]. The CI gate asserts error $< 10^{-4}$.

N	pyMack $c = c_r + ic_i$	$ c - c_{\text{target}} $
64	$0.23753 + 0.00374i$	$\lesssim 10^{-4}$
96	$0.23753 + 0.00374i$	$\lesssim 10^{-5}$
128	$0.23753 + 0.00374i$	$\lesssim 10^{-7}$

2.2 Orszag (1971) Table 5: the full Orr–Sommerfeld spectrum

Reference. Orszag [1], Table 5: the 32 least-stable Orr–Sommerfeld eigenvalues of plane Poiseuille flow at $\alpha = 1$, $R = 10\,000$, each classified symmetric (S) or antisymmetric (A).

Quantity. The full temporal eigenvalue *spectrum*: not only the single least-stable mode of Section 2.1, but the whole branch structure.

Result: agrees. `pyMack`’s dense eigensolver (`validation/test_orszag_full_spectrum.py`) reproduces *all 32* tabulated eigenvalues to a maximum absolute error 7×10^{-6} (median 5×10^{-8}); 32/32 match to $< 10^{-5}$ at $N = 128$. It resolves the A-branch ($c_r \sim 0.96$ downward), the P-branch ($c_r \sim 0.2$ – 0.5), and the vertical S-branch, including the near-degenerate S/A pairs that differ only in the fifth–sixth digit (e.g. modes 2 and 3 at 0.96463 vs. 0.96464). This upgrades the incompressible-core anchor from a single eigenvalue to the full branch structure and exercises the dense full-spectrum path (Figure 1).

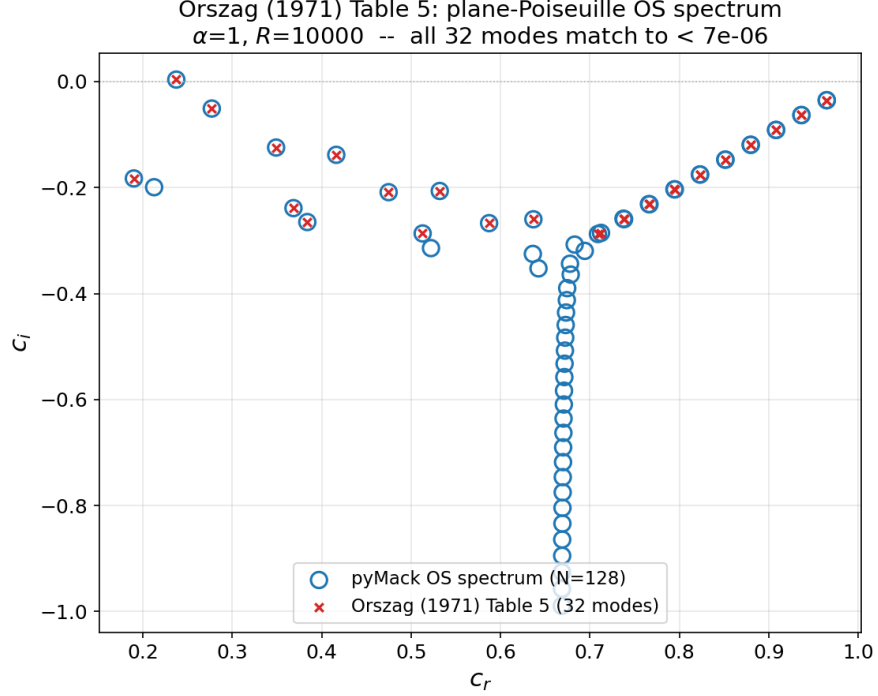


Figure 1: Orszag Table 5 [1]: pyMack’s plane-Poiseuille Orr-Sommerfeld spectrum (circles) over the 32 tabulated eigenvalues (crosses), $\alpha = 1$, $R = 10\,000$. The classic “Y”-shaped spectrum (A-, P-, and S-branches), with every tabulated mode matched to $< 10^{-5}$.

3 Compressible eigenvalue anchors

3.1 Malik (1990): compressible eigenvalue anchors

Reference. Malik [2], the standard tabulated compressible-LST test-case suite. Case 6 (Table IX, the $M = 4.5$ spatial second mode) is the repository’s long-standing anchor, cross-checked digit-for-digit against Xi, Ren & Fu [8] and Hildebrand et al. [9]. Cases 3, 4, 5 (Tables IV/V/VI) and the grid-refinement case of Table X had been logged non-reproducible for want of their dimensional conditions ([9] lists only M and R), until Malik’s own Table I supplied the missing T_0 and wall temperatures (in particular case 4’s *cooled* wall $T_w/T_{\text{adb}} = 0.1$, which earlier attempts had assumed adiabatic), unlocking all four.

Conditions. All cases share $Pr = 0.7$, $\gamma = 1.4$, Sutherland $S = 110.33$ K, L^* length scale. The suite spans $M = 2.5$ –10, temporal (fixed real α , complex ω) and spatial (fixed real ω , complex α) formulations, and adiabatic and cooled walls. Case 3 ($M = 2.5$, oblique $\beta = 0.1$) is a first mode; the rest are second (Mack) modes.

Result: agrees (4 of 5), acceptable (Table X). Table 3. Every real part (α_r or ω_r) is essentially exact ($\leq 0.14\%$), fixing each mode and its phase speed; four of the five *agree*. The one *acceptable* is Table X’s *spatial* α_i (11.5%), which is exactly the documented hypersonic-second-mode inter-method spread (Tumin’s recomputation of case 6’s α_i from Malik’s printed value differs by $\sim 11\%$ in the same way, and the solver of [8] gives -0.002780). Notably the *temporal* $M = 10$ growth rates (cases 4, 5) match to $\sim 2\%$, so the spatial α_i is the more formulation-sensitive quantity.

Case 5 is Malik’s deliberately near-neutral “severe test”: his own SDSP scheme flips ω_i negative at coarse resolution, yet **pyMack** lands on his converged value (Figure 2).

Table 3: The Malik [2] compressible eigenvalue anchors. Spatial cases (6, X) compare the complex wavenumber α ; temporal cases (3–5) compare the complex frequency ω . CI gates: `validation/test_malik1990_case{3,4,5,6}_anchor.py` and `...tableX_anchor.py`.

Case	Conditions	pyMack	Malik	Verdict
6 (Tab. IX)	$M4.5$ adiab., spatial 2nd	$0.253400-0.002490i$	$0.253405-0.002492i$	agrees
3 (Tab. IV)	$M2.5$ oblique, tem- poral 1st	$0.036735+0.000584i$	$0.036734+0.000584i$	agrees
4 (Tab. V)	$M10$ cooled, tempo- ral 2nd	$0.097457+0.001988i$	$0.097484+0.002030i$	agrees
5 (Tab. VI)	$M10$ adiab., tempo- ral 2nd	$0.115862+0.000156i$	$0.115865+0.000153i$	agrees
X (Tab. X)	$M10$ cooled, spatial 2nd	$0.095803-0.002403i$	$0.095933-0.002156i$	acceptable

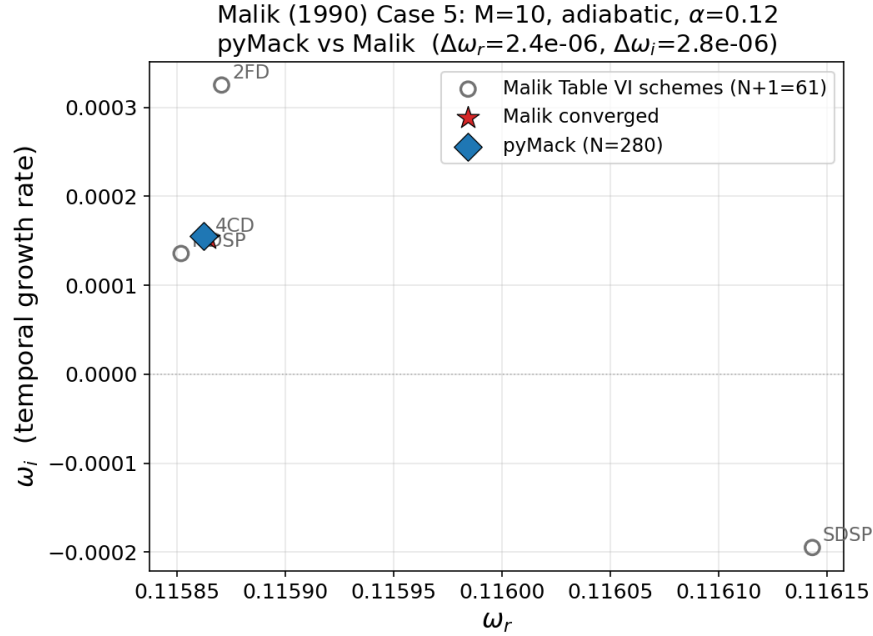


Figure 2: Malik case 5 [2] ($M = 10$ adiabatic, the deliberately near-neutral “severe test”): **pyMack** (diamond) lands on Malik’s converged value (star) and his 4CD/MDSP cluster, while his own 2FD scheme sits high and SDSP flips ω_i negative; **pyMack** reproduces the case Malik built to break solvers.

3.2 Malik (1990) Fig. 4: M10 second-mode eigenfunction

Reference. Malik [2], Fig. 4: the disturbance eigenfunctions of the same $M = 10$ cooled-wall condition as case 4. The comparison uses the temperature eigenfunction $\hat{T}$ (real and imaginary parts), the dominant, cleanly traceable curves; the small velocity and pressure eigenfunctions overlap near the axis and are not reliably digitisable, so they are omitted.

Quantity. The *eigenvector* structure: every other anchor in this document checks *eigenvalues*; this is the *eigenfunction* (mode-shape) check. The complex temperature eigenfunction $\hat{T}(y)$, normalised so that $\hat{T}_r$ peaks at +1 (Malik’s convention, which fixes the eigenvector’s single global phase).

Result: *agrees*. Both the real and imaginary temperature eigenfunctions lie on the digitised Fig. 4 points (Figure 3). **pyMack’s $\hat{T}_r$** traces the near-wall negative lobe, the zero crossing near $y \approx 6$, and the sharp unit peak at $y = 13.3$ ($\approx \delta^*$) against Malik’s ≈ 13 (a 2.3% location match, RMS 0.05 over $0 < y < 20$). **pyMack’s $\hat{T}_i$** reproduces the same double-lobed shape, a positive hump (≈ 0.45) near $y \approx 11$ and a negative dip (≈ -0.5) near $y \approx 15$, to RMS 0.08. Under the *single* global phase that makes $\hat{T}_r$ real at its peak, *both* parts agree, so the temperature eigenvector is validated in full, not merely its peak location. The temperature fluctuation dominates the velocity ($|\hat{T}|/|\hat{u}| \approx 14$), the hallmark of the hypersonic Mack mode.

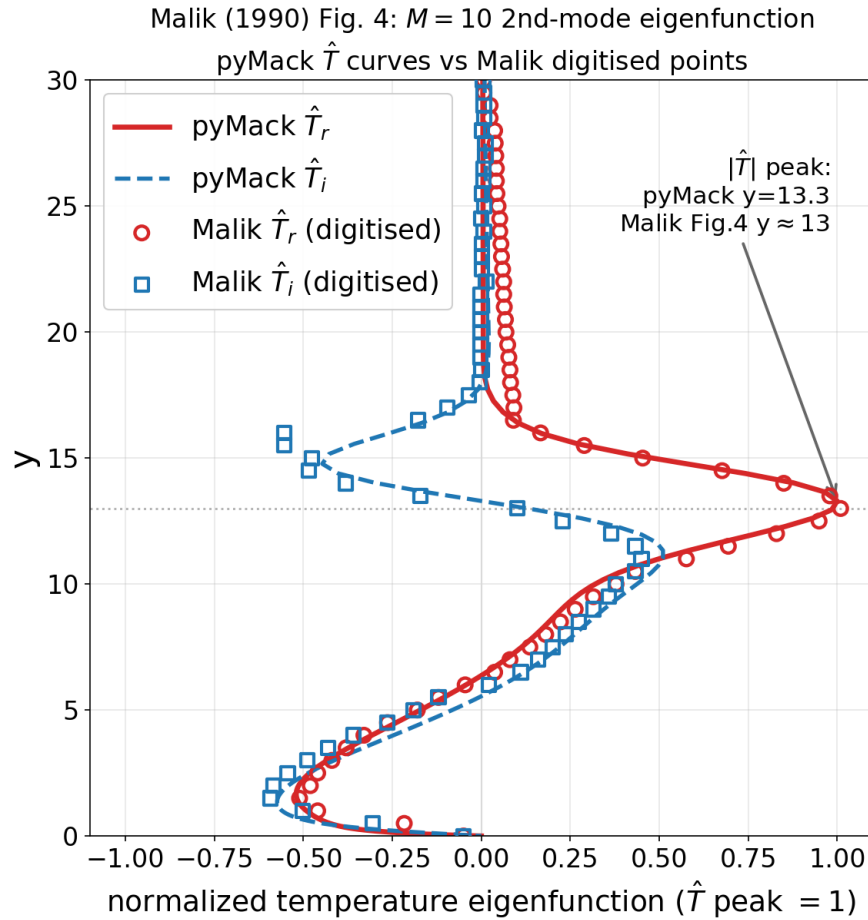


Figure 3: Malik Fig. 4 [2], $M = 10$ second-mode temperature eigenfunction (*agrees*): **pyMack’s $\hat{T}_r$** (solid) and $\hat{T}_i$ (dashed) curves over the digitised points, profile orientation (y vertical), normalised so $\hat{T}_r$ peaks at +1. Both parts match under a single global phase: $\hat{T}_r$ (near-wall lobe, unit peak at $y = 13.3$ vs. ≈ 13 , RMS 0.05) and $\hat{T}_i$ (the +0.45 hump at $y \approx 11$ and -0.5 dip at $y \approx 15$, RMS 0.08).

3.3 Balakumar & Malik (1992): discrete/continuous spectrum topology

Reference. Balakumar & Malik [3]. The paper presents its branches graphically; no tabulated branch eigenvalues are openly recoverable, so this is a qualitative branch-topology check plus a fully specified spatial second-mode cross-check.

Conditions. $M = 4.5$, $T_0 = 311$ K, $Re = 1000$, $\omega = 0.2$, adiabatic (insulated) wall.

Quantity. (a) The topology, the separation of the discrete second mode from the continuous spectrum in the complex- α plane; (b) the discrete spatial eigenvalue α .

Result: *acceptable*. `pyMack`'s full spatial spectrum reproduces the published qualitative topology: the discrete second mode sits at $\alpha = 0.220210 - 0.002787i$ ($c = 0.9082$), matching the published $0.220 - 0.003091i$ (α_r 0.10 %, α_i 9.8 %), and is the unique least-damped root in the second-mode phase-speed band, while a cluster of 93 roots accumulates at the freestream phase speed $c \rightarrow 1$ (the continuous spectrum). A caveat: the separation is in the phase-speed/growth-rate sense, *not* a crisp Euclidean gap: the discretised continuum is dense in α_r , so the nearest other eigenvalue is only 0.019 away (separation ratio 0.09) and 42 roots fall within 10 % of $|\alpha_{\text{disc}}|$. That is the expected discretisation signature of the continuous spectrum crowding the discrete mode, hence *acceptable* (topology correct) rather than *agrees* (which would require a clean Euclidean gap). A companion row, `balakumar_malik1992_via_xirenfu`, treats the same condition purely as a spatial second-mode anchor: `pyMack` $\alpha = 0.2202103 - 0.0027906i$ gives α_r relative error 0.096 % (essentially exact) and α_i relative error 9.7 %, the same ~ 10 % inter-method α_i spread seen on Malik case 6 (Figure 4).

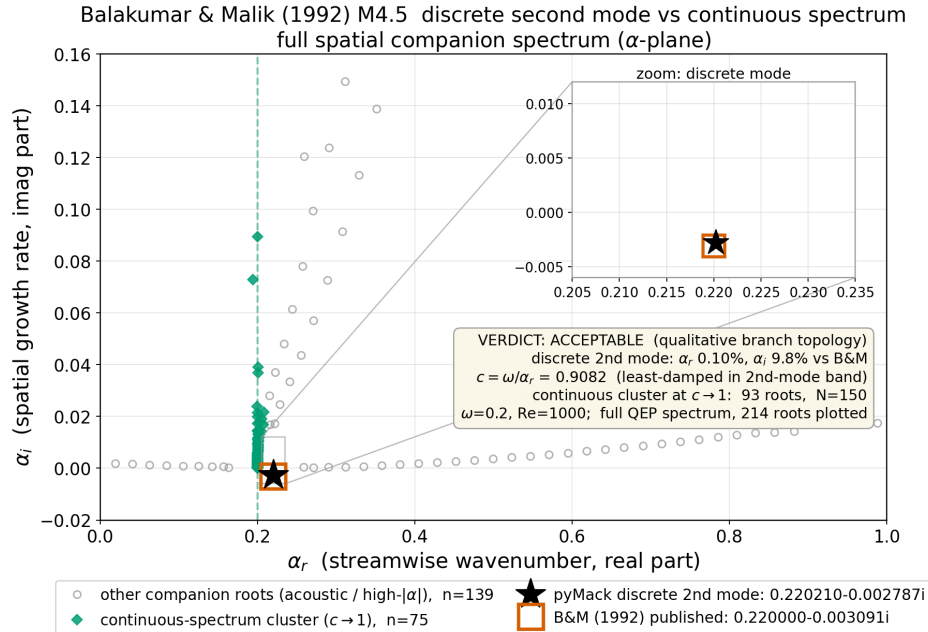


Figure 4: Balakumar & Malik [3], $M = 4.5$: `pyMack`'s complex- α spectrum showing the discrete second mode separated (in phase speed / growth rate) from the continuous-spectrum cluster near $c \rightarrow 1$. Topology correct; the α_r -dense continuum precludes a crisp Euclidean gap.

4 Second mode: curves and families

Having pinned the compressible operator at single points, the following cases test full growth-rate and neutral curves; the second mode is pyMack’s design target.

4.1 Mack (1984) Fig. 10.6: second-mode growth vs. Reynolds number

Reference. Mack [4], Fig. 10.6.

Conditions. Air, adiabatic wall, 2-D wave ($\psi = 0^\circ$), at $M = 4.5, 5.8, 7$, and 10 . Two modelling choices make the comparison defensible: the cold `table_11_1` edge state (resolving a historical $\sim 6\times$ gap that was an edge-temperature error) and a wall-normal domain $y_{\max} \approx 4\delta^*/L^*$ that grows with Mach ($y_{\max} = 40$ at M4.5 up to 140 at M10; a fixed short box starves the thick high-Mach boundary layer).

Quantity. The maximum second-mode temporal growth rate $\omega_{i,\max}$ as a function of R (Mack’s L^* scale, temporal second mode $c_r \approx 0.9$).

Result: *agrees/acceptable*. Median relative error over the overlapping R -stations: **M7: 2.8 %**, **M10: 4.0 %** (at or below the 5 % digitisation floor, verdict *agrees*); **M4.5: 6.0 %**, **M5.8: 8.9 %** (a residual above the floor, verdict *acceptable*). The family agrees at M7 and M10, so the M4.5/M5.8 residuals are local, *not* a monotonic high-Mach under-prediction. The M4.5 and M10 endpoints are shown as representatives (Figures 5 and 6); the M5.8 and M7 overlays exist under the `mack_fig10_6_M58` and `_M70` directories and are covered by the tabulated verdicts, to keep the page count in check. A roll-up case `mack_fig10_6_family` is reserved for the combined M4.5/5.8/7/10 verdict and is marked *pending* in the matrix; the individual panels carry the actual verdicts.

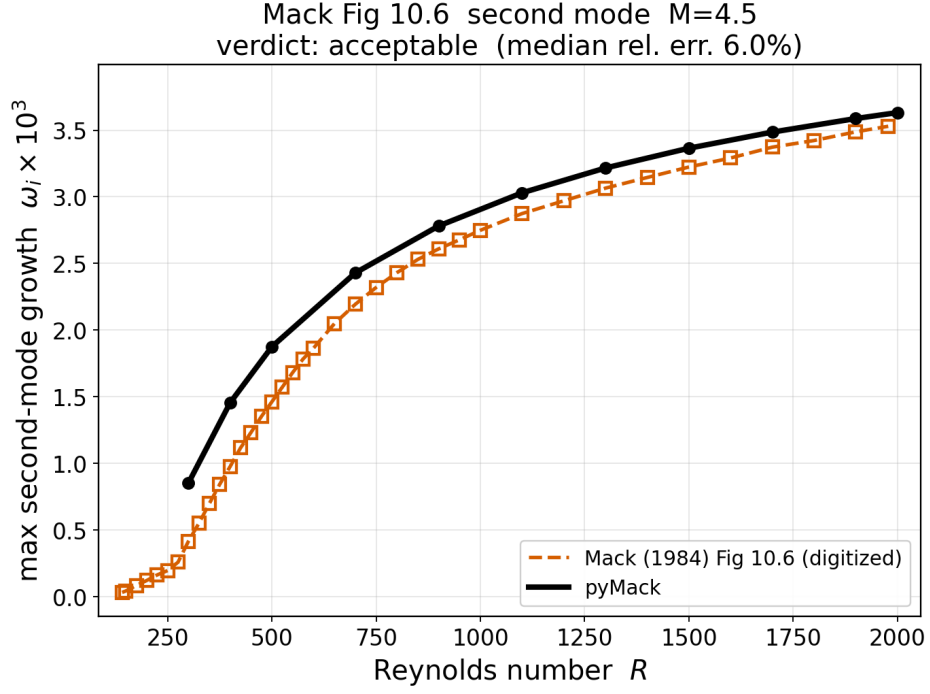


Figure 5: Mack Fig. 10.6 [4], $M = 4.5$: pyMack $\omega_{i,\max}(R)$ (curve) over the digitised points. Median relative error 6.0 %, a small residual above the digitisation floor.

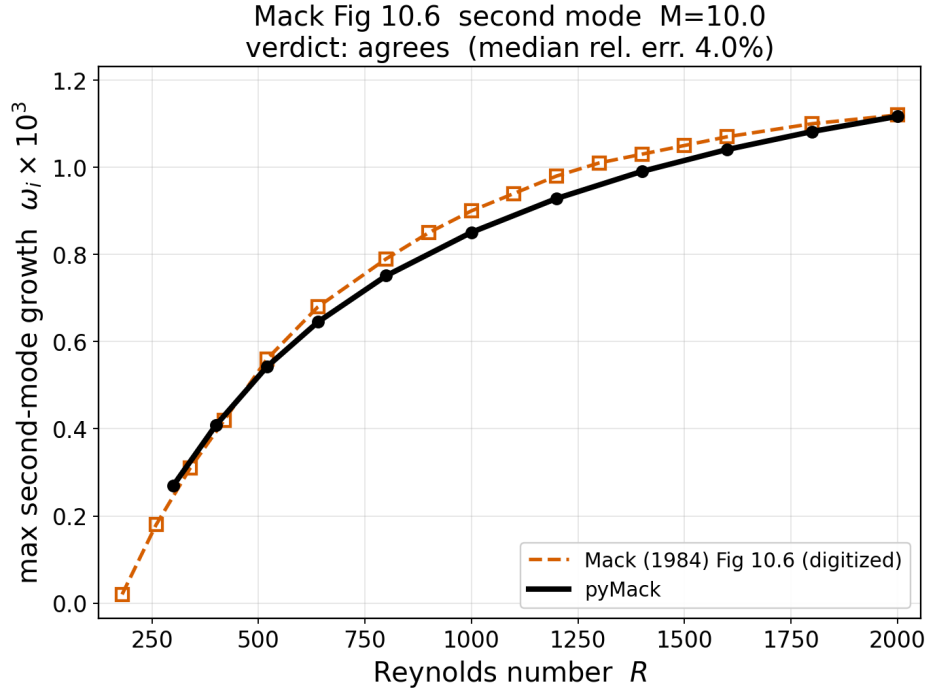


Figure 6: Mack Fig. 10.6 [4], $M = 10$: the same comparison at the top of the Mach range. Median relative error 4.0 % with the domain-converged ($y_{\max} = 140$), cold-edge second mode.

4.2 Egorov et al. (2006): Mach-6 flat-plate second mode

Reference. Egorov, Fedorov & Soudakov [6]: 2-D DNS second-mode growth over a Mach-6 adiabatic flat plate, reported to agree with LST.

Conditions. Perfect gas $\gamma = 1.4$, $Pr = 0.72$, power-law viscosity $\mu/\mu_e = (T/T_e)^{0.7}$; $M_\infty = 6$, $Re_L = 2 \times 10^6$, adiabatic wall; mode-S circular frequency $\omega \approx 200$ (plate-length units) at the synchronisation station $x = 1$ ($R = \sqrt{2 \times 10^6} = 1414$).

Quantity. The second-mode instability band and the most-amplified dimensionless frequency. A caveat: the paper presents the DNS-vs-LST comparison as a growth-rate-vs- x curve with no printed growth-rate digit, so no $\leq 5\%$ numeric anchor exists; the verdict metric is the frequency match.

Result: *acceptable*. `pyMack`'s discrete second mode ($c_r \approx 0.92$ – 0.94) is unstable across $\omega_E \in [140, 260]$ (bracketing the reference forcing $\omega = 200$), most-amplified at $\omega_E \approx 215$ (7.5 % from 200); at exactly $\omega = 200$ the spatial growth is $-\alpha_i = 2.90 \times 10^{-3}$ with $c_r = 0.934$. `pyMack` thus places the M6 second-mode instability squarely in the stated band with the amplified frequency essentially on top of the reference forcing choice (the `egorov2006.m6` overlay is on disk but not embedded here).

4.3 Ma & Zhong (2003) Fig. 15: M4.5 second-mode neutral curve

Reference. Ma & Zhong [7], §6 / Fig. 15: full neutral curves $\omega(R)$ for the first and second mode at $M = 4.5$, with the second-mode branch points quoted in text at $R = 806$ (branch I) and $R = 999.6$ (branch II) for $F = 2.2 \times 10^{-4}$.

Conditions. Air, adiabatic *mean* wall, 2-D ($\psi = 0^\circ$), $M = 4.5$, $T_e = 65.15$ K, unit $Re = 7.2 \times 10^6/\text{m}$, Sutherland $S = 110.33$ K, $Pr = 0.72$, $\gamma = 1.4$; spatial formulation at fixed $F = 2.2 \times 10^{-4}$. The *disturbance* temperature wall condition is isothermal ($\hat{T}|_w = 0$), which the authors state they used for the Fig. 15 neutral curve.

Quantity. The full neutral curve $\omega(R)$ traced by `pyMack`'s $-\alpha_i = 0$ contour, anchored against the two second-mode branch points quoted in the text at $F = 2.2 \times 10^{-4}$.

Result: *agrees*. `pyMack`'s second-mode neutral band reproduces the closed loop of Fig. 15 (nose near $R \approx 280$, upper branch $\omega \approx 0.225$, lower branch $\omega \approx 0.18$) across the full range. At the quoted $F = 2.2 \times 10^{-4}$ station, `pyMack`'s band crosses at $R = 786.6$ (branch I, vs. 806, 2.4 %) and $R = 1009.9$ (branch II, vs. 999.6, 1.0 %), comfortably inside the *agrees* threshold. (Using the adiabatic disturbance condition by mistake gives 831/1033, i.e. 3.1 %/3.4 %; switching to the paper's actual isothermal disturbance condition is a correction, not a fit, and tightens the match.) The first-mode *cutoff* (upper) branch, re-extracted with the discrete-mode selector (freestream decay + $y_{\max}$ -stationarity, replacing the earlier phase-speed-band classifier), tracks Fig. 15's upper branch to $\sim 3.5\%$ median. The low- ω *onset* (lower) branch is *not* recovered: below $\omega \approx 0.03$ the 2-D first mode ($c_r \lesssim 0.45$) delocalises into the slow-acoustic continuous spectrum ($c_r \leq 1 - 1/M = 0.778$), leaving no clean discrete eigenmode. This truncation is consistent across `pyMack`'s spatial and temporal collocation solvers and a QR-shooting variant, and a wave-angle sweep confirms the 2-D mode is the *most* unstable here (so this is a 2-D, not an oblique, first mode). Recovering the onset branch would need a compound-matrix shooting solver robust to the near-continuous-spectrum decay rates; it is filed as an open item in the companion note `work_in_progress.tex` (Figure 7).

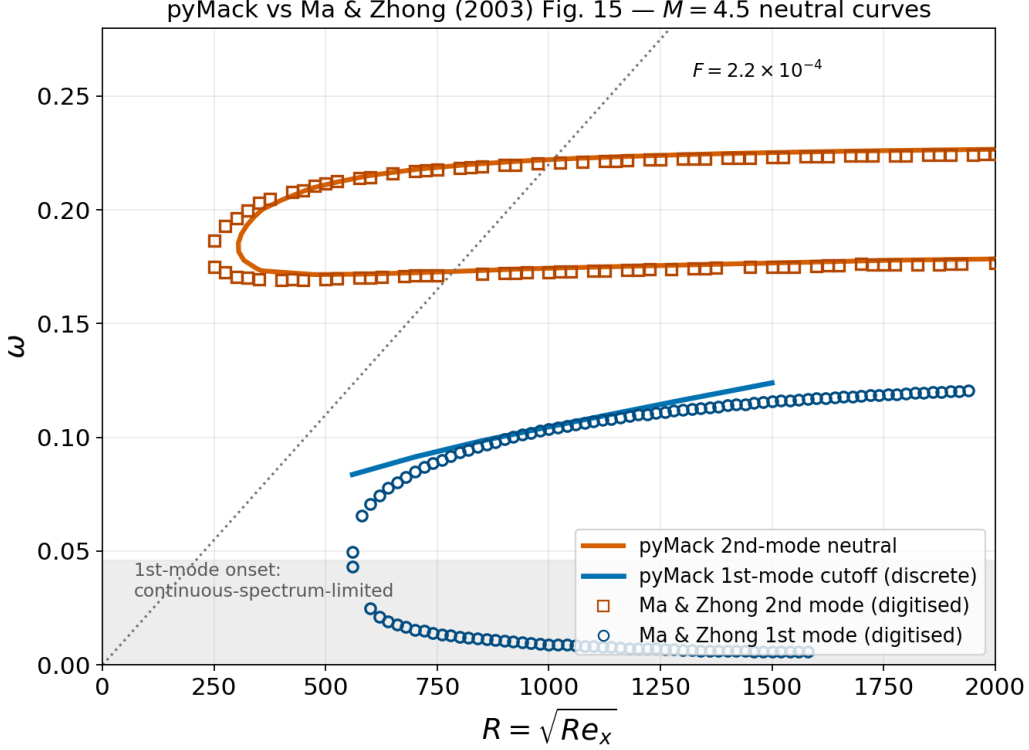


Figure 7: Ma & Zhong Fig. 15 [7], $M = 4.5$ (*agrees*, on the second mode): **pyMack**’s second-mode neutral loop (orange) over the pixel-digitised Fig. 15 points, matching the branch points at $F = 2.2 \times 10^{-4}$ to 2.4% ($R = 786.6$ vs. 806) and 1.0% ($R = 1009.9$ vs. 999.6) under the isothermal disturbance condition. The first-mode *cutoff* branch (blue), re-extracted with the discrete-mode selector, tracks the digitised upper branch to $\sim 3.5\%$; the first-mode *onset* (shaded band) is continuous-spectrum-limited — the long-wavelength mode delocalises into the continuous spectrum, a limit shared by all of **pyMack**’s collocation and QR-shooting methods (see text).

5 First mode: curves and families

The first mode was **pyMack**’s documented weak spot. After a reference-digitisation correction, most of the Mack first-mode figures now agree or are acceptable; the remaining disagreement is narrow and isolated to the $M \approx 2.2$ cases (Fig. 10.1 M2.2 and Fig. 10.3 M2.2), a genuine under-amplification / too-high critical Reynolds number at that Mach rather than a general trend, and is tracked in the companion note `work_in_progress.tex`. This section reports the Mack first-mode figures; the Özgen–Kırcalı neutral-curve family, whose panels contain first-mode branches alongside second-mode ones, is treated separately in Section 6.

5.1 Mack (1984) Fig. 10.1: first-mode neutral loop

Reference. Mack [4], Fig. 10.1, the innermost (complete-equations) neutral loop, re-digitised with both branches plus nose.

Conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$), at $M = 1.6$ and $M = 2.2$; temporal neutral loop ($c_i = 0$) in $(R, F \times 10^4)$.

Quantity. The first-mode neutral loop (lower onset + upper cutoff branches) and the critical Reynolds number.

Result: *acceptable* (M1.6). At M1.6, loop-average median $|\Delta F|/F = 7.7\%$, critical R pyMack ~ 300 vs. Mack ~ 215 , within the acceptable band; the neutral loop closely tracks Mack’s complete-equations loop after the corrected reference (Figure 8). The companion $M = 2.2$ loop *disagrees* (loop-average 54.5 %, critical R pyMack ~ 400 vs. Mack ~ 300 ; a genuine first-mode under-amplification, robust even at the generous end of the $\sim 5\text{--}12\%$ digitisation uncertainty) and is documented in `work_in_progress.tex`.

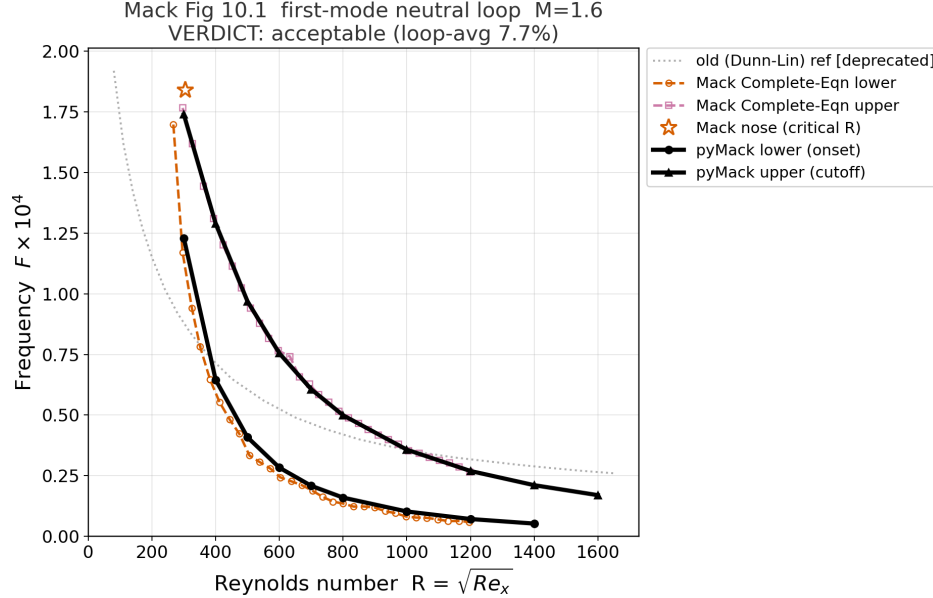


Figure 8: Mack Fig. 10.1 [4], $M = 1.6$ (*acceptable*): pyMack’s first-mode neutral loop closely tracks Mack’s complete-equations loop; loop-average 7.7 %, critical $R \approx 300$ vs. Mack’s ~ 215 .

5.2 Mack (1984) Fig. 10.3: oblique first-mode growth vs. R

Reference. Mack [4], Fig. 10.3.

Conditions. Air (ideal, $Pr = 0.72$), adiabatic wall; oblique waves at $M = 1.3$ ($\psi = 45^\circ$), $M = 2.2$ ($\psi = 45^\circ$), $M = 3.0$ ($\psi = 60^\circ$).

Quantity. Maximum temporal growth $\omega_{i,\max}(R)$ (first mode, optimised over wavenumber α).

Result: *agrees*, with an isolated *disagrees* at M2.2. M1.3: **agrees**, with curve median 1.3 %, and the rigorous Table-10.1 eighth-order anchor matches to 0.91 %. M2.2: **disagrees**, with curve median 32.2 % (convergence-checked: not a $y_{\max}$ or step-count artifact). M3.0: **agrees**, with curve median 3.4 %. Topology is correct in all three (single rising-then-turning first-mode branch); M2.2 is the sole outlier rather than part of a growing-with-Mach trend. The three overlays live under the `mack_fig10_3_m1p3` through `m3p0` directories; the $M = 1.3$ success is embedded below, while the isolated $M = 2.2$ *disagrees* case is documented in `work_in_progress.tex`.

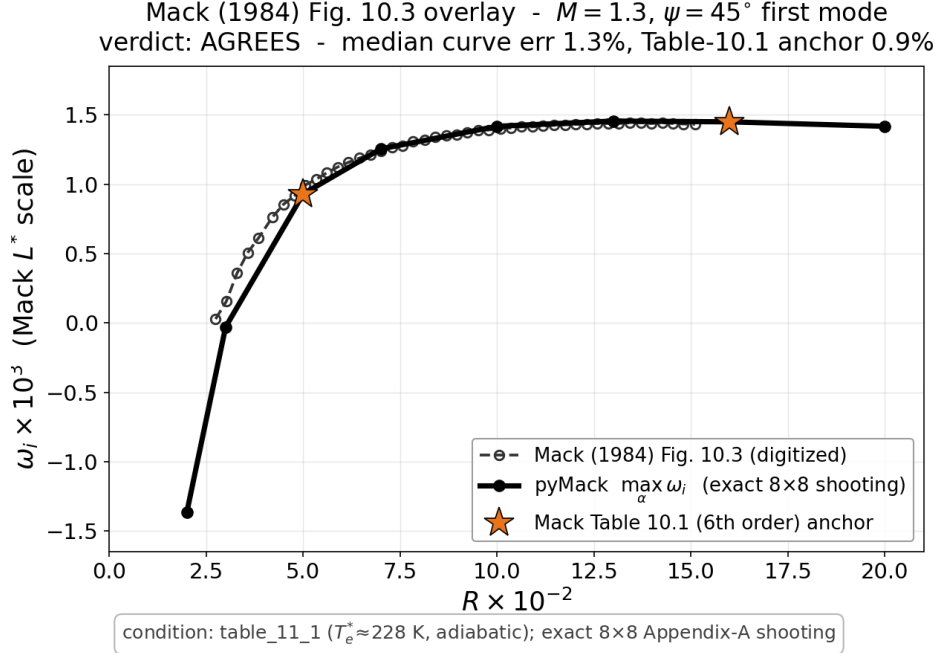


Figure 9: Mack Fig. 10.3 [4], $M = 1.3$, $\psi = 45^\circ$ (*agrees*): $\omega_{i,\max}(R)$ curve median 1.3%; Table-10.1 anchor 0.91 %.

5.3 Mack (1984) Fig. 10.4: oblique first mode at high Mach

Reference. Mack [4], Fig. 10.4.

Conditions. Air, adiabatic wall; the high-Mach first mode is oblique-dominated, so growth is maximised over both α and wave angle ψ ; $M = 4.5, 5.8, 7, 10$ with $y_{\max} \approx 4\delta^*/L^*$.

Quantity. Maximum first-mode temporal growth $\omega_{i,\max}(R)$ optimised over (α, ψ) .

Result: *agrees across M4.5–7, acceptable at M10*. **M4.5:** *agrees*, with curve median 1.4% (peak $\psi \approx 60^\circ$, $c_r \approx 0.83$). **M5.8:** *agrees* (2.8%), **M7:** *agrees* (1.7%), **M10:** *acceptable* (6.1%). A confidence note: these were computed with the reduced dense-EVP path (`solve_temporal_compressible_3d`), not the exact-shooting path used for the convergence-checked Fig. 10.3 cases, and no explicit exact-shooting cross-check was recorded; the domain used the second-mode-calibrated $\sim 4\delta^*/L^*$ rule (the first mode reaches further from the wall). The entire family now agrees or is acceptable; there is no first-mode under-amplification trend at high Mach in this figure; the small M10 residual is the only case above the digitisation floor. A `mack_fig10_4_family` aggregate is *pending*. M4.5 and M10 are embedded (Figures 10 and 11); the M5.8 and M7 overlays exist under the `mack_fig10_4_M58` and `_M70` directories.

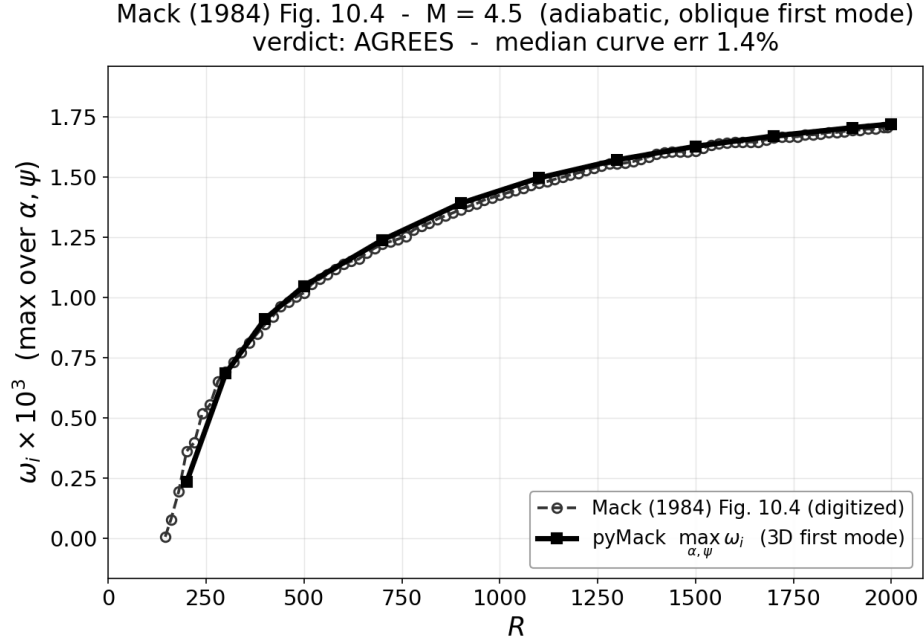


Figure 10: Mack Fig. 10.4 [4], $M = 4.5$ (*agrees*): oblique first mode optimised over (α, ψ) , curve median 1.4%.

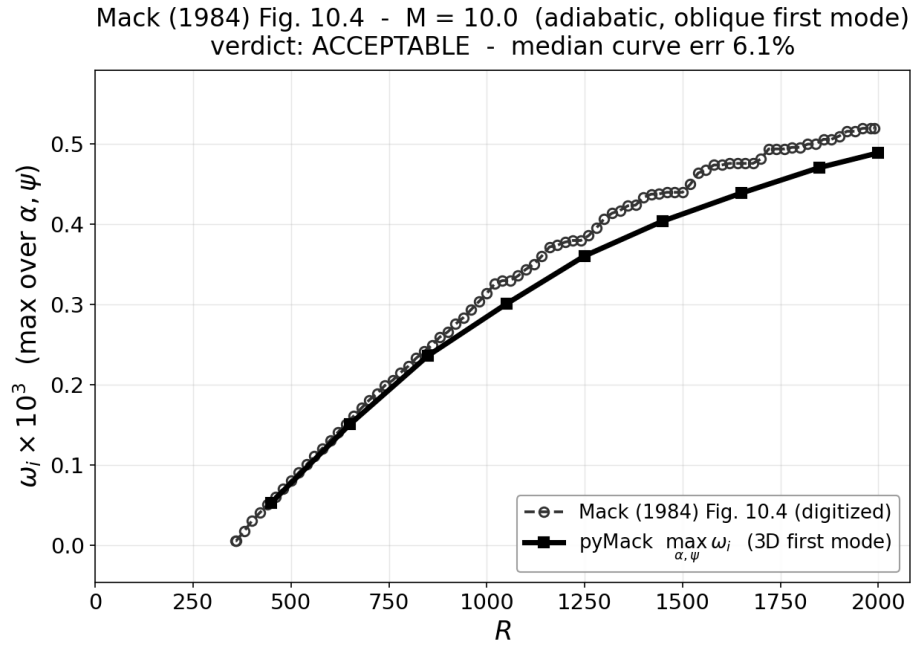


Figure 11: Mack Fig. 10.4 [4], $M = 10$ (*acceptable*): curve median 6.1%, a small residual above the digitisation floor at the top of the Mach range (lower-confidence magnitude, see text).

6 Mixed first and second mode: the Özgen–Kırcalı neutral-curve family

Fig. 3 of Özgen & Kırcalı [5] presents, for each Mach number from 2 to 10, a single neutral-curve figure in the (Re_L, α_L) plane. At $M = 2$ and 3 the figure contains the first-mode lobe alone; from $M \approx 4$ the second (Mack) mode enters, and each panel then spans *both* instability families; at $M = 6$, for example, four branches: second-mode upper and lower, first-mode cutoff, and first-mode onset. Because a single reproduced figure tests both modes at once, these cases are classified neither under the first nor the second mode but as a mixed family (`verification/mixed_mode/ozgen_fig3/`), and they are reported last, drawing on both preceding sections. The per-panel `verdict.json` records `mode: first` for M2/M3 and `mode: first+second` for M4–M10.

Reference. Özgen & Kırcalı [5], Fig. 3 ($M = 2, 3, 4, 6, 7, 8, 10$).

Conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$); temporal neutral contour ($c_i = 0$) in the (Re_L, α_L) plane.

Quantity. The neutral-curve branches $\alpha_L(Re_L)$: all branches present in each panel, first *and* second mode.

Result: *agrees/acceptable*. After the references were re-digitised correctly and a discrete-mode extractor (eigenfunction decay + $y_{\max}$ -stationarity, the two acceptance tests of the companion *Numerical Methods* document) replaced the earlier c_r -band classifier that had excluded the genuine first mode, the picture is:

- **M4: agrees** (Figure 12). Complete two-lobe match: both second-mode branches at the digitisation floor (0 % upper / 0 % lower), first-mode cutoff 2 % over ~ 85 % coverage; only the deep low- α first-mode onset tracks at 32 %.
- **M2, M3, M6: acceptable** (Figures 13 and 14). M2 traces both first-mode branches over the full R -range by eigenvalue continuation (lower 7 %, upper 8 %); M3 first-mode cutoff 6 % where resolvable (onset continuous-spectrum-blocked); M6 four branches with second-mode upper 2 %, lower 3 %, first-mode cutoff 9 %, first-mode onset 50 %.
- **M7, M8, M10: acceptable (topology caveat).** The dominant second-mode cutoff branch agrees (2–6 %), but `pyMack` predicts a *connected* first–second-mode band where the reference shows two lobes separated by a stable gap. This is a documented connected-vs-separated topology difference (inter-mode over-prediction at high Mach), not a second-mode-curve error.

A separate growth-contour case `ozgen_fig3_lobes` **disagrees** ($c_i = 0.004$ lobe median 18.8 %, and the interior $c_i = 0.012$ M2 lobe is absent because `pyMack`’s peak M2 growth $c_{i,\max} = 0.0109 < 0.012$), which is the same open-lobe / under-amplification discrepancy. A combined all-Mach overlay exists at `figures/ozgen_fig3_overlay.png`. To hold the page count, only M2, M4, and M6 are embedded below; the M3/M7/M8/M10 overlays exist under `verification/mixed_mode/ozgen_fig3/M*` and are covered by the tabulated verdicts.

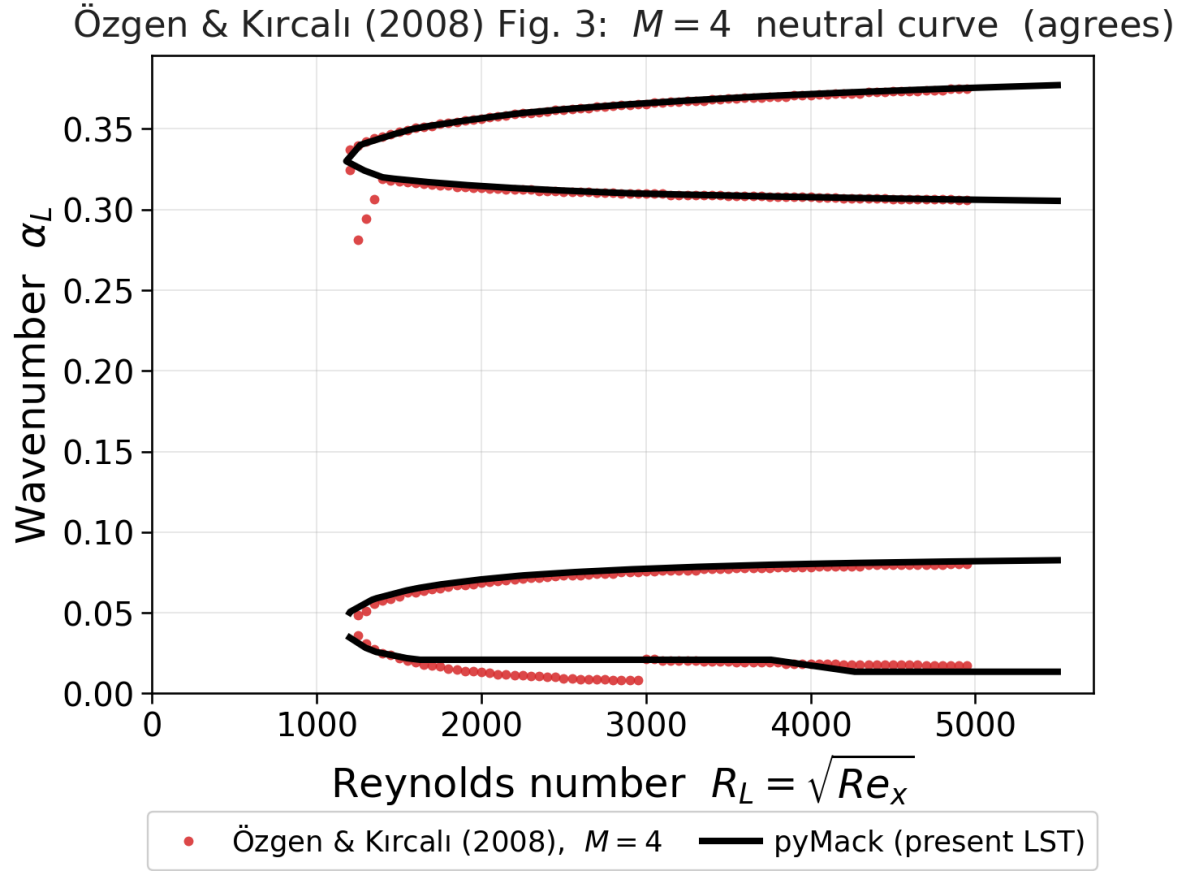


Figure 12: Özgen & Kırçalı Fig. 3 [5], $M = 4$ (*agrees*): pyMack reproduces the entire two-lobe neutral structure: both second-mode branches at the digitisation floor and the first-mode cutoff to 2%.

Özgen & Kırçalı (2008) Fig. 3: $M = 2$ neutral curve (acceptable)

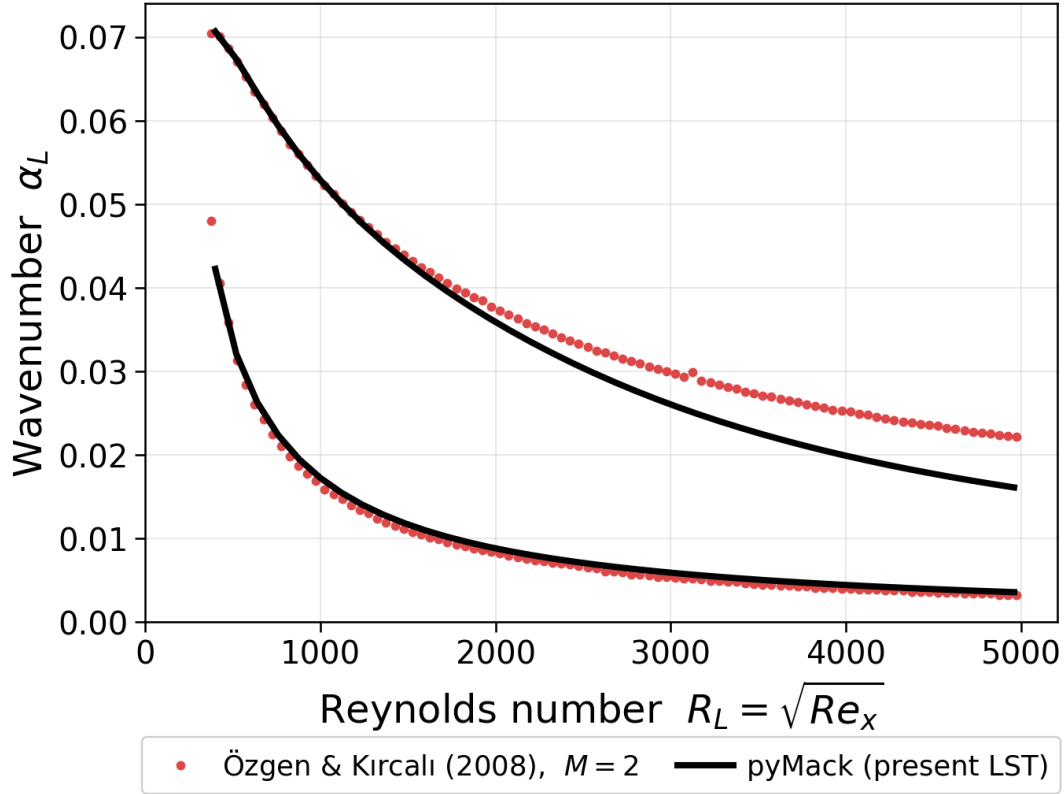


Figure 13: Özgen & Kırçalı Fig. 3 [5], $M = 2$ (*acceptable*; first mode only at this Mach): both branches over the full R -range by eigenvalue continuation (lower 7%, upper 8%).

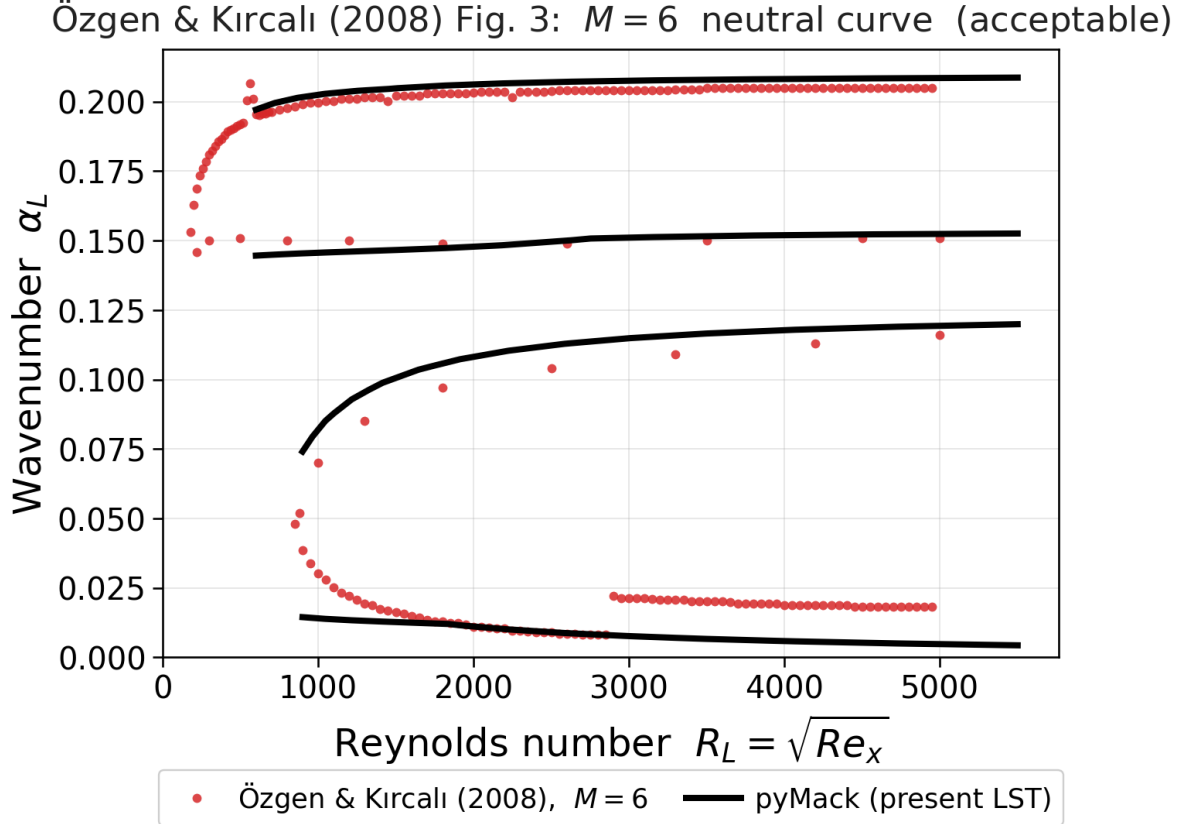


Figure 14: Özgen & Kırçalı Fig. 3 [5], $M = 6$ (*acceptable*; both modes in one panel): four branches, with second-mode upper/lower 2/3 % and first-mode cutoff 9 % agree; only the deep low- α first-mode onset (50 %) is extraction-limited.

Cases that are not settled validations (the isolated $M \approx 2.2$ first-mode disagreements, the sharp-cone N -factor with its bounded magnitude deficit, the private Mach 5.35 code-to-code benchmark, and the Mach 6 pipeline demonstration) are collected in the companion note `work_in_progress.tex`.

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